

ass8

April 6, 2025

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.naive_bayes import GaussianNB, BernoulliNB
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix, \
    classification_report
```

```
[10]: # Load dataset properly
df = pd.read_csv("C:/Users/cool/Desktop/ML ass/pima-indians-diabetes.data.
    ↪csv", header=None)

# Now assign the correct column names
df.columns = ['X1', 'X2', 'X3', 'X4', 'X5', 'X6', 'X7', 'X8', 'X9']

# Remove the first row if it contains old column names
df = df.drop(index=0)

# Reset index
df.reset_index(drop=True, inplace=True)

# Display
df.head()
```

```
[10]:  X1  X2  X3  X4  X5  X6  X7  X8  X9
0  6  148  72  35   0  33.6  0.627  50  1
1  1   85  66  29   0  26.6  0.351  31  0
2  8  183  64   0   0  23.3  0.672  32  1
3  1   89  66  23  94  28.1  0.167  21  0
4  0  137  40  35 168  43.1  2.288  33  1
```

```
[12]: # Check for missing values
df.isnull().sum()
```

```
[12]: X1    0
      X2    0
      X3    0
      X4    0
      X5    0
      X6    0
      X7    0
      X8    0
      X9    0
      dtype: int64
```

```
[14]: # Split features and target
      X = df.drop('X9', axis=1)
      y = df['X9']

      # Train-test split
      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
      ↪random_state=42)

      # Feature scaling (optional for Naive Bayes, but improves results sometimes)
      scaler = StandardScaler()
      X_train_scaled = scaler.fit_transform(X_train)
      X_test_scaled = scaler.transform(X_test)
```

```
[16]: # Gaussian Naive Bayes
      gnb = GaussianNB()
      gnb.fit(X_train_scaled, y_train)
      y_pred_gnb = gnb.predict(X_test_scaled)

      # Bernoulli Naive Bayes
      bnb = BernoulliNB()
      bnb.fit(X_train_scaled, y_train)
      y_pred_bnb = bnb.predict(X_test_scaled)
```

```
[24]: # Step 5 - Evaluation for GaussianNB
      from sklearn.metrics import accuracy_score, f1_score

      # Evaluate
      acc_gnb = accuracy_score(y_test, y_pred_gnb)
      f1_gnb = f1_score(y_test, y_pred_gnb, average='weighted') # Add average
      ↪parameter

      # Print results
      print("---- Gaussian Naive Bayes ----")
      print(f"Accuracy: {acc_gnb:.2f}")
      print(f"F1 Score: {f1_gnb:.2f}")
```

```
--- Gaussian Naive Bayes ---  
Accuracy: 0.77  
F1 Score: 0.77
```

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[ ]:
```